We acknowledge the use of OpenAI’s ChatGPT as a supportive tool in drafting and refining portions of the text.

THE UNIVERSITY OF BRITISH COLUMBIA
Faculty of Forestry

UFOR 225: Urban Ecosystem Mapping and Monitoring

Winter Term 2 2026/27

ACKNOWLEDGMENT

UBC’s Point Grey Campus is located on the traditional, ancestral, and unceded territory of the xwm■θkw■y■■m (Musqueam) people. The land it is situated on has always been a place of learning for the Musqueam people, who for millennia have passed on their culture, history, and traditions from one generation to the next on this site.

Title	Urban Ecosystem Mapping and Monitoring (UFOR 225)
Credits	3
Term	WT2: in-person classes on Fridays from 10:00 to 13:00. Some activities are scheduled outdoors on c Classroom: Hugh Dempster Pavilion (DMP) | Floor 3 | Room 305
Grading	Numeric (0-100%)

Teaching Team

Instructor: Dr. Maya Henderson

I look forward to meeting you as your instructor for this course. My research focuses on urban ecosystem resilience, climate adaptation, and environmental stewardship in cities. My work explores how urban forests, green infrastructure, and biodiversity initiatives can support healthier and more sustainable communities.

E-mail: maya.henderson@ubc.ca
Office hour: Friday 2:00 – 3:00 pm

GTA: Jordan Kim

I am currently completing a Master of Forestry degree focused on urban biodiversity and ecosystem monitoring. I previously worked on campus sustainability projects involving tree inventories and environmental data collection.

E-mail: jordan.kim@student.ubc.ca
Office hour: Wednesday 1:00 – 2:00 pm

COURSE DESCRIPTION

Urban Forestry 225 provides students with an overview of urban ecosystem mapping, monitoring, and assessment. The course introduces methods used to evaluate urban green infrastructure, biodiversity, and environmental conditions within cities. Students will work with environmental datasets, field observations, mapping tools, and ecosystem indicators to understand how monitoring supports planning and management decisions.

LEARNING OUTCOMES

By the end of the course, students will be able to:

• Explain the importance of urban ecosystem monitoring and assessment.
• Compare different inventory and mapping approaches.
• Identify environmental indicators used in urban ecosystem studies.
• Design and implement a basic ecosystem monitoring exercise.
• Synthesize findings into planning recommendations and presentations.

COURSE STRUCTURE

The course extends over a full term with a combination of lectures, field activities, guest speakers, group discussions, and applied assignments. Students will participate in field exercises and complete group-based monitoring projects using environmental assessment tools.

URBAN ECOSYSTEM MONITORING
– Importance of environmental monitoring
– Mapping and inventory techniques
– Environmental indicators and analysis

URBAN BIODIVERSITY AND ASSESSMENT
– Ecosystem service assessment
– Observation and mapping techniques
– Environmental data visualization

URBAN PLANNING AND STEWARDSHIP
– Urban ecosystem planning
– Integrating assessment into decision-making
– Communication of findings and recommendations

ASSESSMENT, EVALUATION AND GRADING

Participation (10%)
Short assignments (20%)
Assignment 1: Ecosystem inventory (30%)
Assignment 2: Environmental assessment (30%)
Individual presentation (10%)

Activity	Weight (%)
Participation in class	10
Short assignments and assessments	20
Assignment 1: Ecosystem Inventory	30
Assignment 2: Environmental Assessment	30
Individual presentation	10
Total	100

IMPORTANT DATES AND DEADLINES

Activity	Date
First day of class	Jan 8
Fieldwork demonstration	Jan 22
Start of field inventory	Jan 29
Assignment 1 submission	Feb 26
Experience mapping fieldwork	Mar 12
Individual presentations	Apr 2
Assignment 2 submission	Apr 16

COURSE SCHEDULE

Topic	In-person seminars	Asynchronous or out-of-class activities
Week 1: Getting started (Jan 4-8)	– Introduction to the course – Community agreement – Introduction to campus greening strategies	– Nature-based bio – Study group establishment
MODULE I		

Week 2: Ecosystem inventory design (Jan 11-15)	– Lecture: inventory methods – Lecture: ecosystem indicators	– Expectations and motivations
Week 3: Introduction to fieldwork (Jan 18-22)	– Demonstration in the field – Tree measurement techniques	– Tree walk
Week 4: Inventory fieldwork (Jan 25-29)	– Fieldwork in groups	– Complete sample plots
MODULE II		
Week 8: Ecosystem assessment (Feb 22-26)	– Ecosystem service frameworks – Intro to environmental assessment tools	– Submit Assignment 1
MODULE III		
Week 12: Planning and stewardship (March 22-26)	– Urban ecosystem planning – Peer feedback sessions	– Complete infographics

REQUIRED AND OPTIONAL READINGS

Week	Required readings	Optional readings
1	No readings	McPherson, C. 2018. Urban ecology and resilience.
2	Nielsen, A.B. 2014. Review of urban tree inventory methods.	Morgenroth, J. 2017. Measuring and monitoring urban trees.
3	Magarik, Y.A. 2021. Tree measurement methods.	Urban monitoring resource guide.
4	No readings	

EQUIPMENT

- Raincoat and rain pants for outdoor field work
- Cellphones with mapping applications installed
- Laptop computer for group work and assignments
- Students will receive field gear and are expected to maintain it in good condition

ACADEMIC INTEGRITY

Students are expected to uphold the highest standards of academic integrity. Generative AI tools may be used to support idea generation and studying with proper acknowledgement.

OTHER UNIVERSITY POLICIES

UBC provides resources to support student learning, accessibility, wellbeing, and respectful communication within the academic community.